AP CALCULUS AB

Unit 1 • Worksheet 1-1A • Answer key

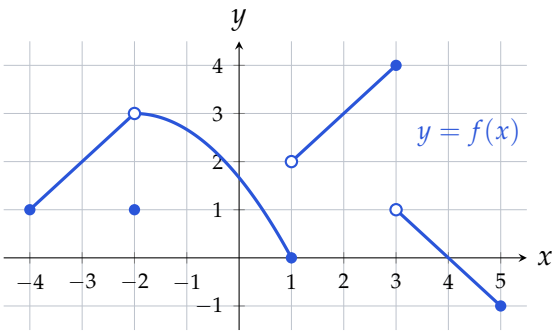
Worksheet 1-1A • Limits to a Point on a Graph

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Questions 1 to 8 refer to the graph of f on $[-4, 5]$.



Question 1

NO CALCULATOR

What is $\lim_{x \to -2} f(x)$?

(A) 1

(B) 3

(C) It does not exist.

(D) 0

Solution

Answer: (B). From both sides the graph approaches the open circle at height 3.

Question 2

NO CALCULATOR

What is $f(-2)$?

(A) 3

(B) 0

(C) 1

(D) It does not exist.

Solution

Answer: (C). $f(-2)$ is the height of the filled dot at $x = -2$.

Question 3

NO CALCULATOR

What is $\lim_{x \to 1^-} f(x)$?

(A) 0
(B) 2
(C) It does not exist.
(D) 1

Solution
Answer: (A). From the left, the curved piece comes down to the filled dot $(1, 0)$.

Question 4

NO CALCULATOR

What is $\lim_{x \to 1} f(x)$?

(A) 0
(B) 2
(C) 1
(D) It does not exist.

Solution
Answer: (D). Left limit 0, right limit 2: different, so the limit does not exist.

Question 5

NO CALCULATOR

What is $\lim_{x \to 3^+} f(x)$?

(A) 4
(B) 1
(C) It does not exist.
(D) 3

Solution
Answer: (B). From the right, the graph approaches the open circle at $(3, 1)$.

Question 6

NO CALCULATOR

What is $f(3)$?

(A) 1
(B) It does not exist.
(C) 4
(D) 3

Solution
Answer: (C). The filled dot at $x = 3$ is at height 4.

Question 7

NO CALCULATOR

For which values of x in $(-4, 5)$ does $\lim f(x)$ fail to exist?

(A) $x = 1$ and $x = 3$
(B) $x = 1$ only
(C) $x = -2$ and $x = 1$
(D) $x = -2$, $x = 1$ and $x = 3$

Solution
Answer: (A). At $x = 1$ and $x = 3$ the one-sided limits differ. At $x = -2$ the limit exists (3), even though $f(-2) = 1$.

Question 8

NO CALCULATOR

Which statement is true?

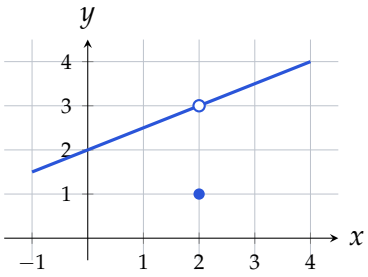
(A) $\lim_{x \to -2} f(x) = f(-2)$
(B) $\lim_{x \to 3^-} f(x) = 1$
(C) $\lim_{x \to 4} f(x) = 0$
(D) $f(1) = 2$

Solution
Answer: (C). On the last piece, the height at $x = 4$ is 0.

Question 9 (Free response)

NO CALCULATOR

(a) Using the graph of f, find $\lim_{x \to -3} f(x)$, $\lim_{x \to 3^-} f(x)$ and $\lim_{x \to 3} f(x)$.
(b) On the grid, sketch a function g with $\lim_{x \to 2} g(x) = 3$ but $g(2) = 1$.
(c) Explain the difference between $\lim_{x \to c} f(x)$ and $f(c)$.



Solution
(a) 2; 4; does not exist ($4 \neq 1$). (b) Example: a line through $(2, 3)$ with an open circle there, and a filled dot at $(2, 1)$. (c) The limit is the height the graph approaches near c; $f(c)$ is the actual height at c. They can be different, and either one can fail to exist.